HONG WOON YUN

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OBJECTIVE

To contribute to the successful execution of the AI Capstone Design project

PERSONAL PROFILE

Dedicated researcher specializing in the fabrication and characterization of neuromorphic semiconductor devices based on advanced materials, including 2D materials (TMDs) and oxide semiconductors. My core expertise lies in developing floating-gate-based memristors to implement high-performance synaptic and neuronal functions. I am committed to advancing large-scale integrated neuromorphic systems that bridge the gap between device-level innovation and hardware-level neural network implementation

EDUCATION

Ph.D in AI System Engineering September 2024 - Present
Sungkyunkwan University (SKKU), Suwon, South Korea.
Overall GPA: 4.36 / 4.5 (27 credits completed)

Master of Science in AI System Engineering September 2022 - August 2024
Sungkyunkwan University (SKKU), Suwon, South Korea.
GPA: 4.06 / 4.5
Thesis: [Two-terminal Au Nanoparticle Floating-gate Memristor for Neuromorphic Computing]

Bachelor of Engineering March 2016 - August 2022
Sungkyunkwan University (SKKU), Suwon, South Korea.
Dual Major: Systems Management Engineering & Electronic and Electrical Engineering
Overall GPA: 3.69 / 4.5
Major GPA (Electronic and Electrical Engineering): 4.12 / 4.5

CAPSTONE DESIGN

Project Title : Multi-terminal Short-term Floating-gate Memristor for Implementing Convolutional Neural Network

Project Description : The proposed multi-terminal short-term floating-gate memristor enables not only column-wise one-dimensional pooling based on temporal summation, but also row-wise pooling based on spatial summation. As a result, simultaneous row- and column-wise pooling can be achieved, allowing two-dimensional pooling operation. To evaluate the effectiveness of the two-dimensional pooling operation, learning simulations were performed using the results obtained from both one-dimensional and two-dimensional pooling. The simulation results showed that two-dimensional pooling achieved higher learning accuracy than one-dimensional pooling.

Research Papers for Reference : Nat. Commun. 15, 9147 (2024), Nat. Commun. 15, 2044 (2024), Nat. Commun. 14, 3070 (2023), Small 2503877. (2025)

Project Achievements
Software Registration : 1
Title : MNIST Patch-Based Device Output Conversion and Classification Learning Simulator with Selectable Multi-Reservoir Sizes
Patent Application : 1
Title : Fully Integrated On-Chip Neuromorphic Device for Spiking Neural Network Implementation Without External Auxiliary Circuits